

Bhadra Korukonda

AI Engineer

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Detail-oriented Computer Science student with practical experience in AI and Machine Learning. Developed graph-based knowledge retrieval systems (GRAIL-LM) integrating GNNs and LLMs, and NLP-driven resume ranking models. Skilled in Python, data analysis, scalable deployment, and model performance optimization techniques.

SKILLS

Programming Languages: Python, Java, JavaScript

Libraries/Frameworks: PyTorch, scikit-learn, NumPy, Flask, React

Tools/Platforms: FastAPI, Flask, REST APIs, Git, Docker, Kubernetes, Prometheus, Grafana

Databases: PostgreSQL, MongoDB, SQL

PROFESSIONAL EXPERIENCE

Research Centre Imarat(RCI), DRDO

Data Scientist Intern |

June 2024 – Aug 2024

- Automated Git workflows with Python, cutting version control overhead by **30%**.
- Developed statechart validation tools using OOP and tests, boosting accuracy by **25%**.
- Streamlined pipelines with SQL and Git CI, accelerating debugging and deployment.

IIT Hyderabad

Research Intern | Python, PyTorch Geometric, Pandas, Geopy, GRU/LSTM, GNNs (GCN, GAT, EGAT) May 2025 – Aug 2025

- Implemented a hybrid Edge-Aware Graph Attention Network (EGAT-GRU) to forecast PM2.5 levels across Delhi using spatio-temporal air quality data.
- Encoded wind direction and geodesic distance as edge features, enabling wind-informed pollutant transport learning and dynamic spatial attention.
- Achieved state-of-the-art performance (RMSE = 16.72, MAE = 10.05), outperforming GAT, GCN, and physics-based baselines for real-time pollution prediction.

PROJECTS

GRAIL-LM | FastAPI, LangChain, Neo4j, Docker, Ollama (LLaMA3)

- Developed an end-to-end Graph-RAG system integrating Large Language Models with Knowledge Graphs using FastAPI, LangChain, and Neo4j for explainable multi-hop question answering.
- Achieved ~25-30% improvement in factual accuracy and sub-2 s response latency by grounding LLM outputs in graph-structured data and deploying a containerized API-Streamlit pipeline.
- Engineered a production-ready AI stack (FastAPI + PyTorch Geometric + Docker) demonstrating scalable graph reasoning, LLM integration, and real-time visualization of reasoning paths.

RelFuse MVP | Pytorch, DGL, Transformers, GraphSAGE, FastAPI

- Developed a hybrid relational-feature fusion model combining GNN embeddings with transformer attention for structured data reasoning.
- Conducted feature ablation and hyperparameter tuning, achieving +18 % classification accuracy on benchmark graph datasets.
- Built modular inference APIs to serve predictions and visualizations, bridging research models with deployable analytics tools.

Automated Resume Ranking System using NLP and MCDM | Python, Flask, spaCy, scikit-learn, PyMCDM, Tesseract OCR

- Developed an AI-driven resume evaluation system integrating Natural Language Processing (NLP) and Multi-Criteria Decision-Making (MCDM) for automated, explainable candidate ranking.
- Extracted and analyzed resume data (skills, education, experience, projects) using spaCy, NLTK, and Tesseract OCR to generate structured features for scoring.
- Implemented TOPSIS/AHP algorithms to compute weighted candidate scores, improving shortlisting efficiency by ~40% and reducing human bias.
- Built a Flask-based API and dashboard enabling HR users to upload resumes, adjust evaluation weights, and visualize ranked results in real time.
- Ensured modular design for scalability and transparency, allowing independent updates to NLP or MCDM components.

EDUCATION

Bachelor of Technology (CS Engg), Woxsen University, GPA: 3.0/4.0

Aug 2022 – July 2026

Narayana Jr College(SSC), Hyderabad, Percentage: 87.1%

June 2020– May 2022

Chinmaya Vidyalaya(CBSE),Hyderabad Percentage: 90%

May 2008 – June 2020